HARSHITA POOJARY

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EDUCATION

University of Southern California, Los Angeles, CA.
Master of Science - Computer Science **GPA: 3.81/4.00**

University of Mumbai, Mumbai, India.
Bachelor of Engineering - Computer Engineering **GPA: 3.83/4.00**

TECHNICAL SKILLS

Languages: Python, TypeScript, Go, C++, Java, JavaScript
Frameworks: PyTorch, TensorFlow, LangChain, Node.js, FastAPI, AngularJS
AI/ML & Infra: LLM fine-tuning, model serving, vector DBs, retrieval, CI/CD, GPUs (T4/A100), distributed inference
Platforms & Tools: Kubernetes, Docker, Redis, MySQL, MongoDB, ElasticSearch, Milvus, GitLab CI, Azure, AWS, Grafana, Prometheus

WORK EXPERIENCE

AI Intern **Jul 2025 – Present**
Deep Defense Solutions (Remote) *USA*

- Designed and shipped **agentic LLM pipelines architecture** & **orchestration** frameworks using **Python** and **LangChain** with vector stores, automating routing and verification workflows and reducing manual load for community ops teams.
- Implemented safety guardrails including prompt-injection mitigation, schema-validated outputs, and **LLM-as-judge** evaluation with human review, with fallback strategies and human handoff to ensure reliable, policy-compliant behavior.

Data Science Intern **Jun 2024 – Aug 2024**
Zero-True (Remote) *New York, USA*

- **Enabled data-driven analysis** by delivering 10 SQL- and Python-based interactive dashboards, allowing non-technical stakeholders to explore datasets and monitor system behavior dynamically.
- Improved product usability by implementing 5 reusable **UI components (Vuetify)**, increasing accessibility and consistency across notebook-based workflows.
- Strengthened cross-functional collaboration by incorporating stakeholder and customer feedback into feature iterations, supporting requirement clarification and delivery alignment.

Machine Learning Engineer **Dec 2021 – Jul 2023**
Reliance Jio *Mumbai, India*

- Reduced face-registration latency by **40%** by deploying RetinaFace with **TensorRT** FP16 on **T4/A100 GPUs**, optimizing cost-latency-accuracy tradeoff suites for real-time distributed inference.
- Increased deployment velocity by **50%** by automating **CI/CD pipelines** with **Python, Node.js, Docker, Kubernetes, GitLab, and Azure services**, embedding health checks, secure coding practices, and monitoring in production.
- Achieved **90%** spoof-detection accuracy and scaled to 20+ sites by fine-tuning **ResNet-32** and establishing offline evals for accuracy, false positives, and drift, integrated with production monitoring.
- Improved engineering quality by mentoring 4 interns through code reviews and validation workflows, reinforcing production readiness and coding standards.

Software Engineer **Jun 2019 – Dec 2021**
Reliance Jio *Mumbai, India*

- Architected Node.js microservices and content-delivery **REST APIs** for JioNews/JioGate (100M+ users) on Kubernetes with **MongoDB, MySQL, Redis,** and **Elasticsearch**, implementing sharded data flows, OAuth authentication and RBAC-based secure access control.
- Built large-scale distributed ingestion pipelines for real-time data processing with **Kafka, Node.js, Go, SQL, Kubernetes**, improving throughput **35%** across multi-device platforms with high availability.
- Reduced API downtime by **15%** by designing and operating automated integration tests, observability, and alerting systems **(Grafana, Prometheus, Kibana**), improving issue detection, escalation, and incident triage.
- Improved execution predictability by collaborating with product and engineering stakeholders to align API designs with UX requirements, identify dependencies, estimate timelines, and support iterative feature delivery.

Software Engineer **Jan 2019 – Jun 2019**
TaksyKraft (Remote) *Hyderabad, India*

- Developed real-time engagement prediction APIs in **Python** and **SQL**, improving targeting accuracy and increasing client ROI by **10%**.
- Shortened reporting cycles by **40%** by building **ETL pipelines** and **Tableau dashboards**, accelerating enterprise decision-making.

PROJECTS

Unified Semantic Space for Multimodal Retrieval (PyTorch, Python, LanceDB)

- Designed a multimodal retrieval framework aligning **BridgeTower** text–image embeddings for video-based Q&A, integrating **LanceDB** for sub-second vector search and deploying an interactive **Streamlit** interface.

Analyzing NLP Techniques on Low-Resource Languages (PyTorch)

- Performed **idiom and metaphor classification** for low-resource **Konkani** using **BERT-based transformers** and applied **attention-head pruning**, maintaining model accuracy while reducing computational overhead for efficient deployment.

Comparison of Model Performance with Knowledge Distillation and Quantization (PyTorch, Python)

- Fine-tuned **LLaMA-3-8B** with **4-bit** quantization and **LoRA**-based logit distillation, reaching **76.9% accuracy** on **GSM8K (8-shot)** while reducing inference cost compared to the original 8B model.